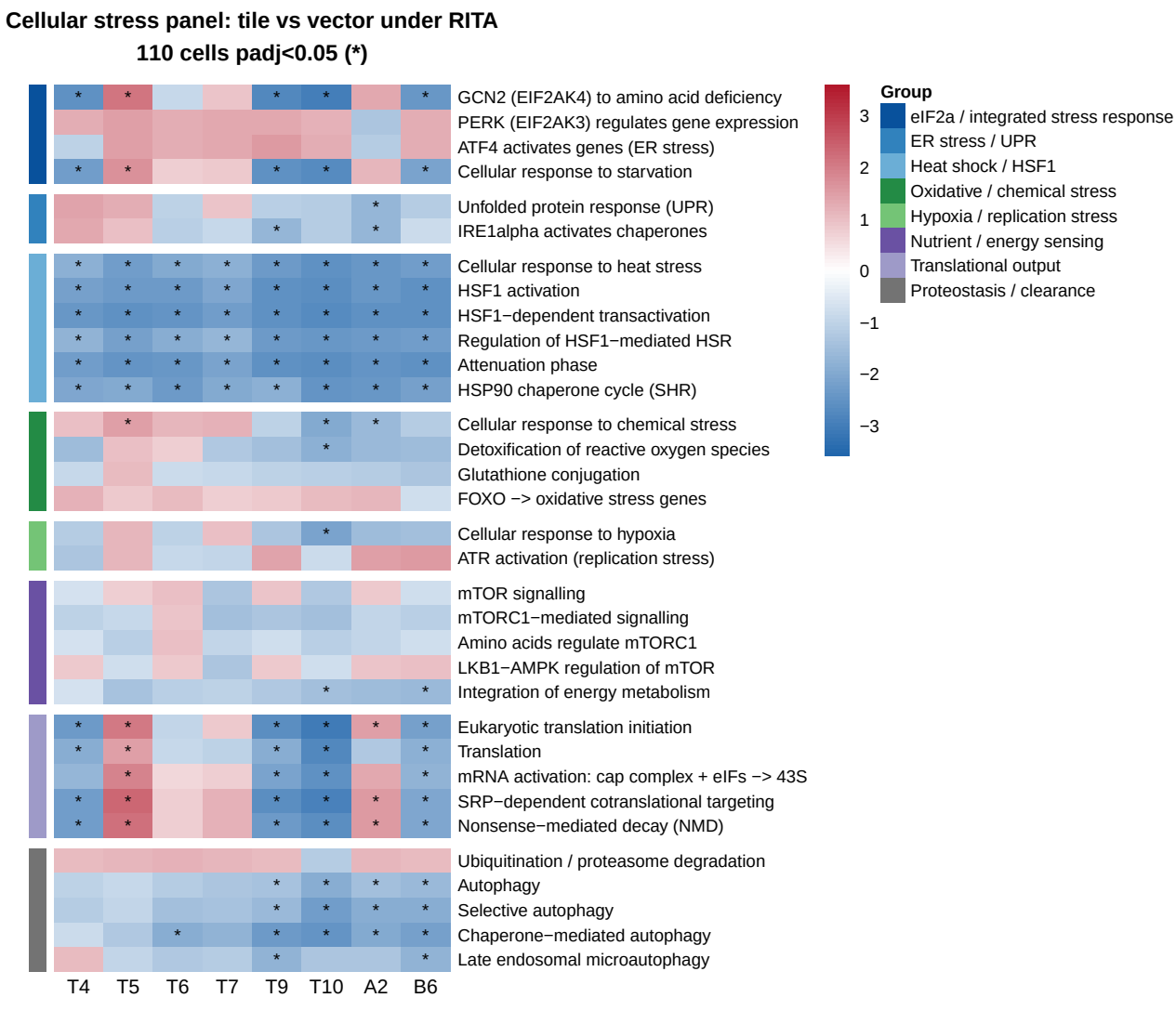
Cellular stress panel: tile vs vector under RITA
110 cells padj<0.05 (*)
GCN2 (EIF2AK4) to amino acid deficiency
PERK (EIF2AK3) regulates gene expression
ATF4 activates genes (ER stress)
Cellular response to starvation
Unfolded protein response (UPR)
IRE1alpha activates chaperones
Cellular response to heat stress
HSF1 activation
HSF1−dependent transactivation
Regulation of HSF1−mediated HSR
Attenuation phase
HSP90 chaperone cycle (SHR)
Cellular response to chemical stress
Detoxification of reactive oxygen species
Glutathione conjugation
FOXO −> oxidative stress genes
Cellular response to hypoxia
ATR activation (replication stress)
mTOR signalling
mTORC1−mediated signalling
Amino acids regulate mTORC1
LKB1−AMPK regulation of mTOR
Integration of energy metabolism
Eukaryotic translation initiation
Translation
mRNA activation: cap complex + eIFs −> 43S
SRP−dependent cotranslational targeting
Nonsense−mediated decay (NMD)
Ubiquitination / proteasome degradation
Autophagy
Selective autophagy
Chaperone−mediated autophagy
Late endosomal microautophagy
T4
T5
T6
T7
T9
T10
A2
B6
3
2
1
0
−1
−2
−3
Group
eIF2a / integrated stress response
ER stress / UPR
Heat shock / HSF1
Oxidative / chemical stress
Hypoxia / replication stress
Nutrient / energy sensing
Translational output
Proteostasis / clearance